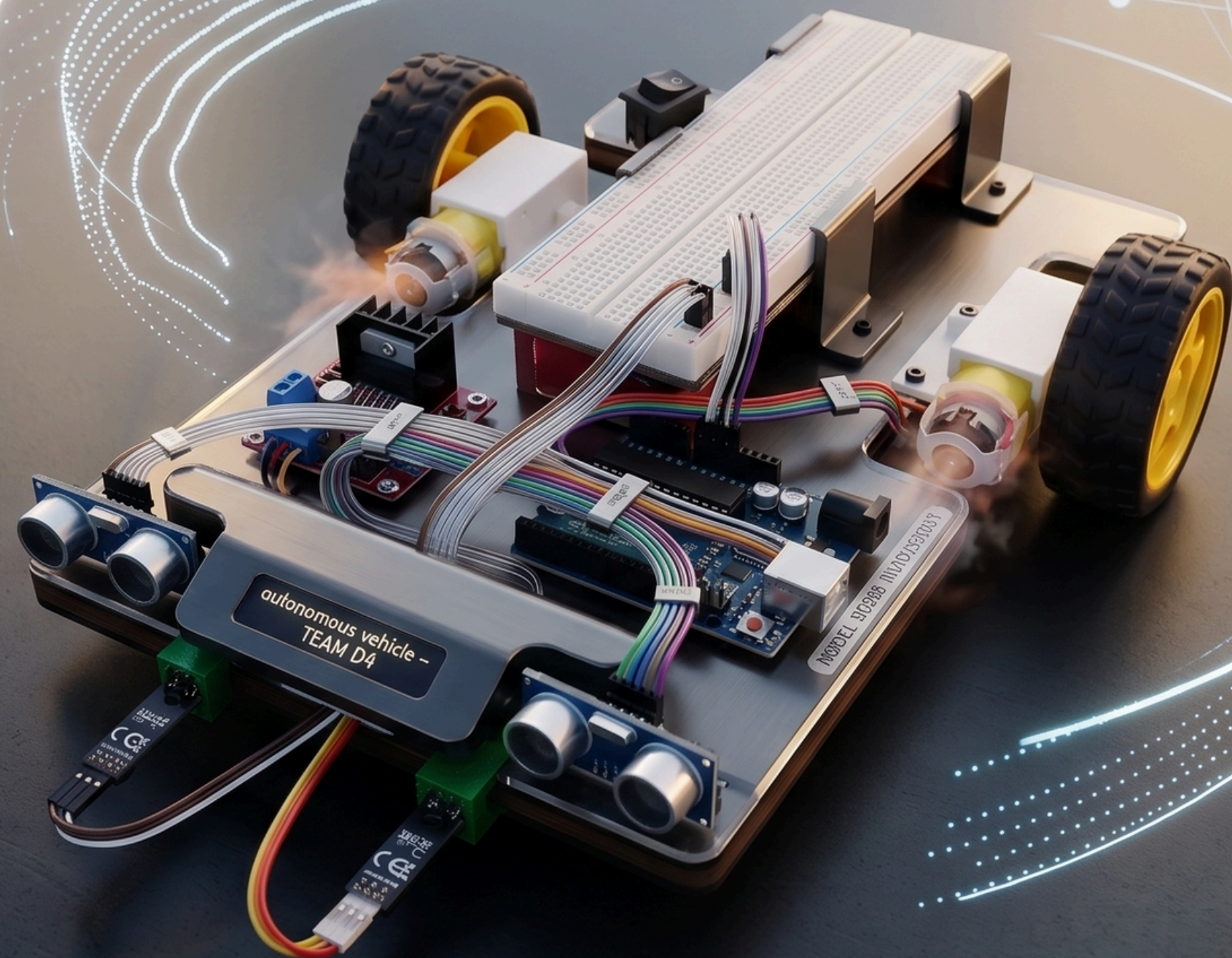




#1



DESIGN AND IMPLEMENTATION OF AN AUTONOMOUS LINE- FOLLOWING AND OBSTACLE- AVOIDING ROBOT

Presented by ;

- **Dodly Forbi**
- **Ye-Htet_Kyaw**
- **Gabriel Tchakunte**
- **Abdelmalek Abraham**

**ELE Systems
Engineering
SS2026**

PRESENTATION OUTLINE

- **Project objectives & Requirements**
- **System Engineering View**
- **Hardware and Mechanical Design**
- **Software and Control Logic**
- **UML Diagrams**
- **Testing and Results**
- **Conclusion**
- **Future Work**

PROJECT OBJECTIVES & REQUIREMENTS

OBJECTIVES

- **Line Following**

Follow straight and curved tracks

- **Obstacle Detection and varied behaviour ;**

Perform alternating responses such as obstacle dodging and 180° turn

CONSTRAINTS

- **Arduino Uno only**

- **Kit Components only including ;**

- **A pair of Ultrasonic & IR sensors, L298N motor driver , 2 motors, breadboard , screws, wires and a custom chassis**

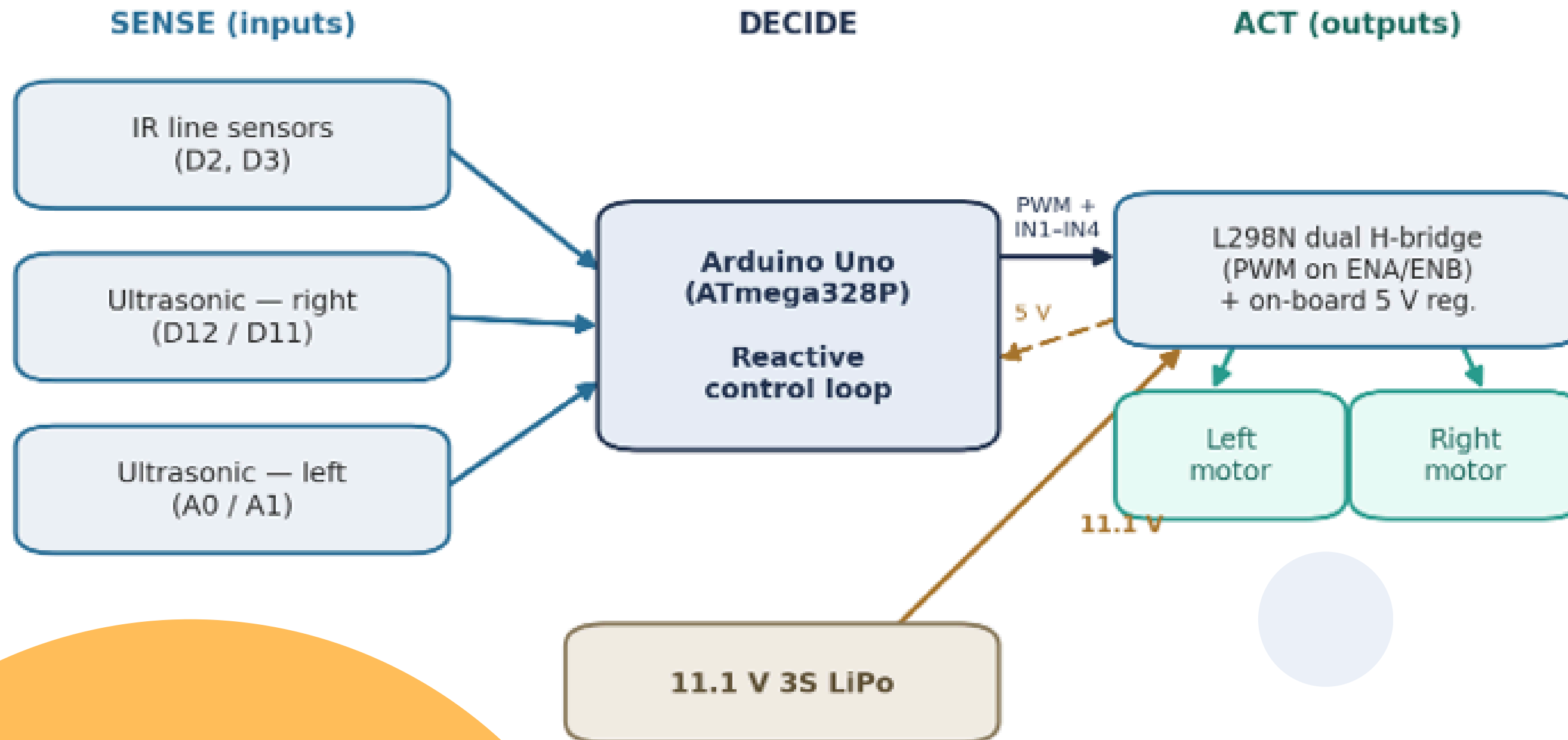
- **Custom track**

- **Environment**

SYSTEM ENGINEERING VIEW

Slide 3

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Inputs

- 2× IR sensors
- 2× HC-SR04 sensors

Controller

- Arduino Uno

Outputs

- L298N motor driver
- Differential drive motors

Power

- 11.1V LiPo

HARDWARE AND MECHANICAL DESIGN

Presented by ;

- **Dodly Forbi**
- **Ye-Htet_Kyaw**
- **Gabriel Tchakunte**
- **Abdelmalek Abraham**

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Engineering
SS2026**

HARDWARE IMPLEMENTATION (1/5)

Component	Purpose	Holder Designed
Arduino Uno	Main controller	No
L298N	Motor driver	No
IR Sensors	Line detection	Yes
Ultrasonic Sensors	Obstacle detection	Yes
LiPo Battery	Power supply	Yes
DC Motors	Locomotion	Yes



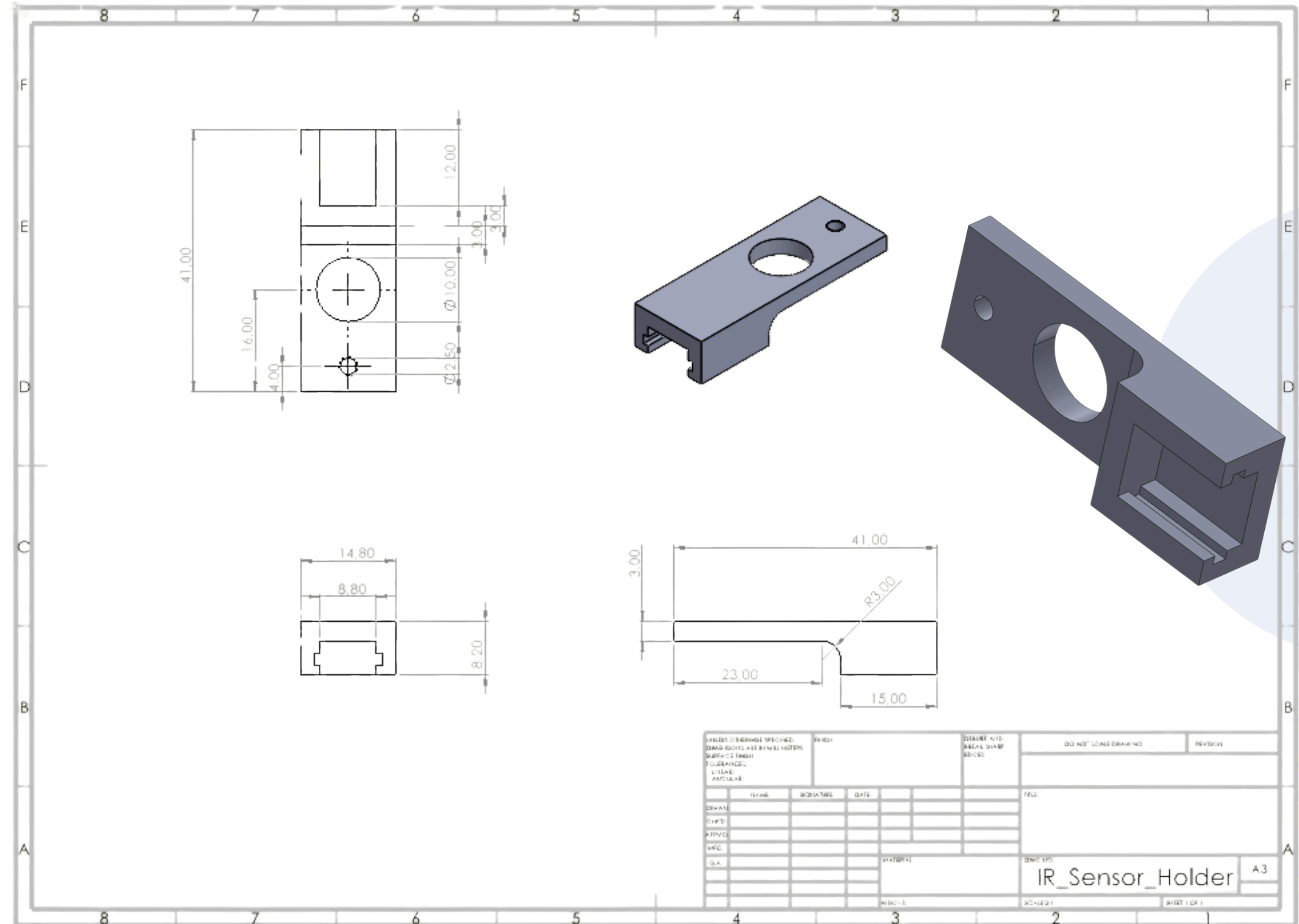
Mounts and Holders were designed on SolidWorks and then printed

Slide 6

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- 1

Ye Htet Tchew

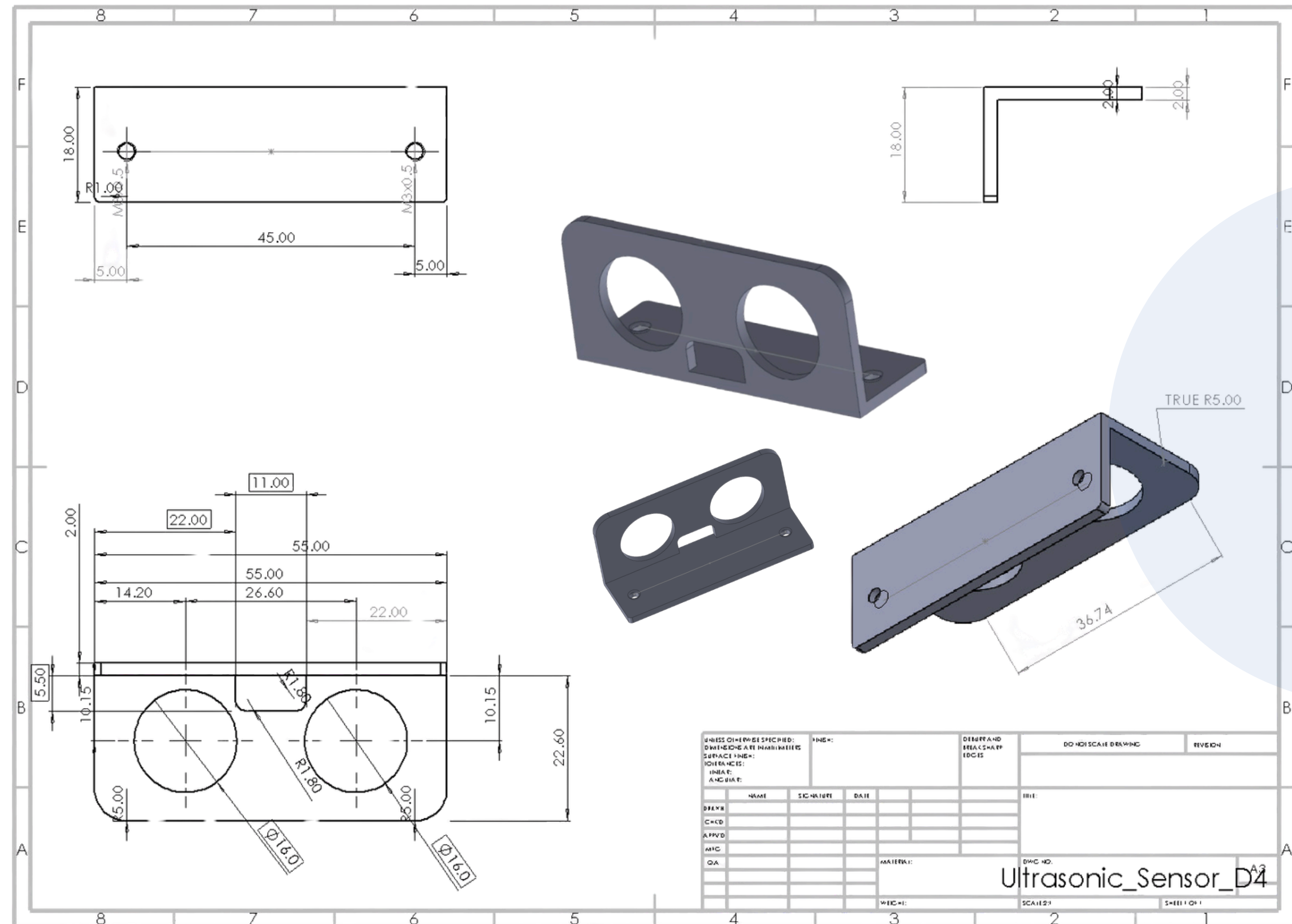


Slide 7

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- # 2

Gabriel
Tchakunte

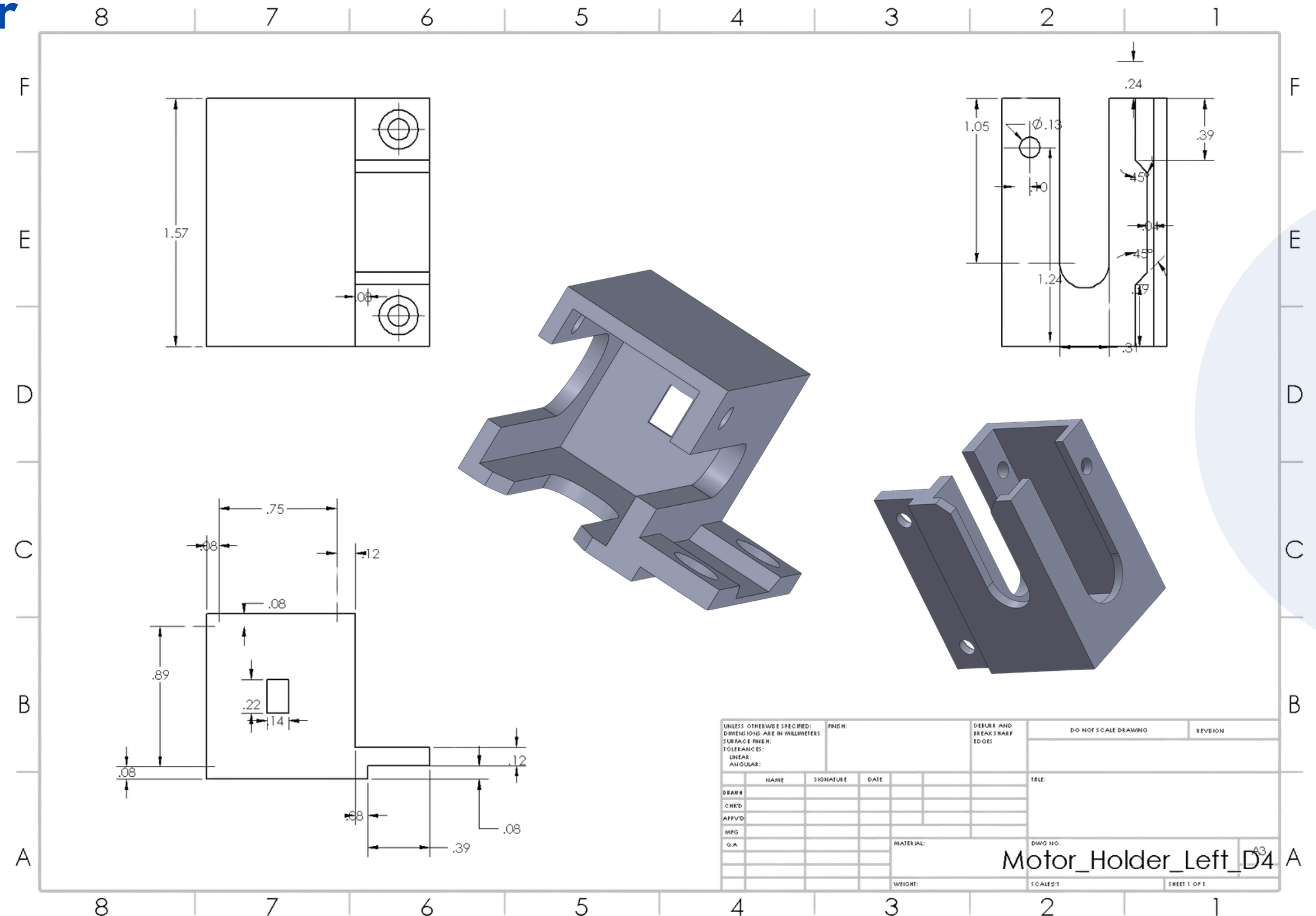


Slide 8

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3

Abdelmalek
Abraham

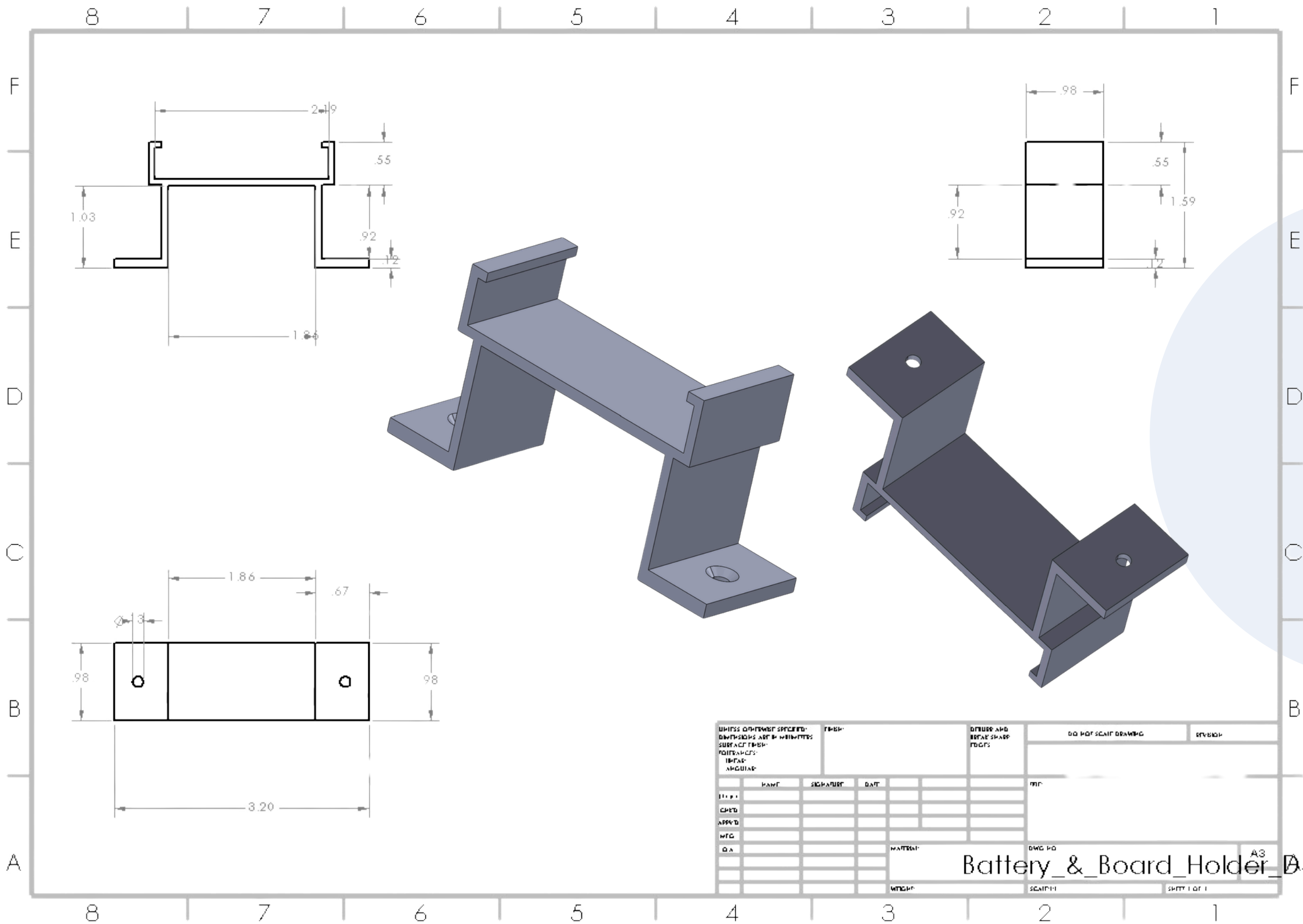


HARDWARE IMPLEMENTATION (5/5)

- Battery/Breadboard Holder

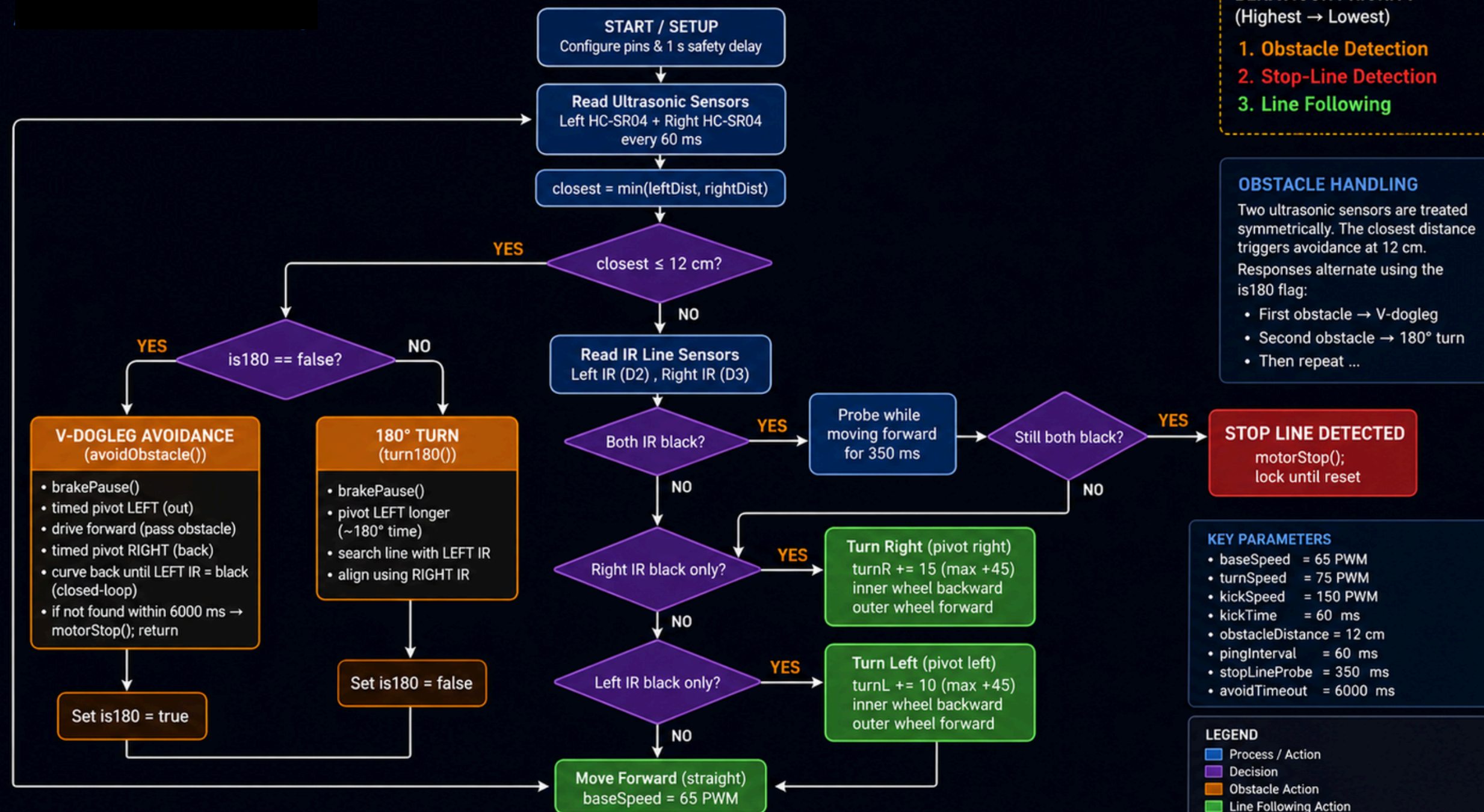
4

Dodly Forbi



SOFTWARE & CONTROL LOGIC (1/3)

Slide 10



```
if (!is180)
    avoidObstacle();
else
    turn180();
```

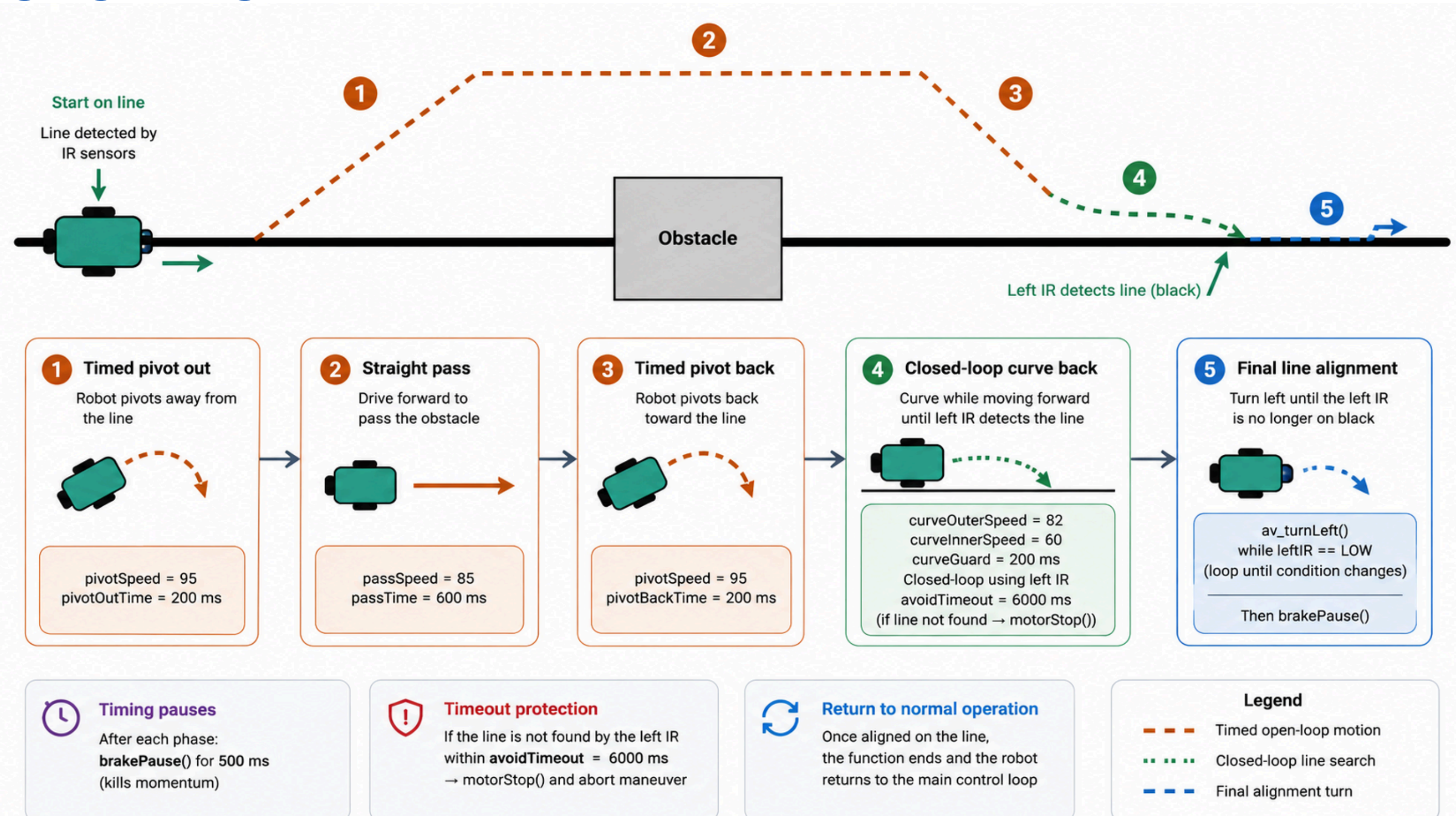
- Two ultrasonic sensors checked every 60 ms
- Closest distance triggers obstacle response at 12 cm
- is180 alternates V-dogleg and 180° turn

SOFTWARE & CONTROL LOGIC (2/3)

Slide 11



• V dogleg dodge

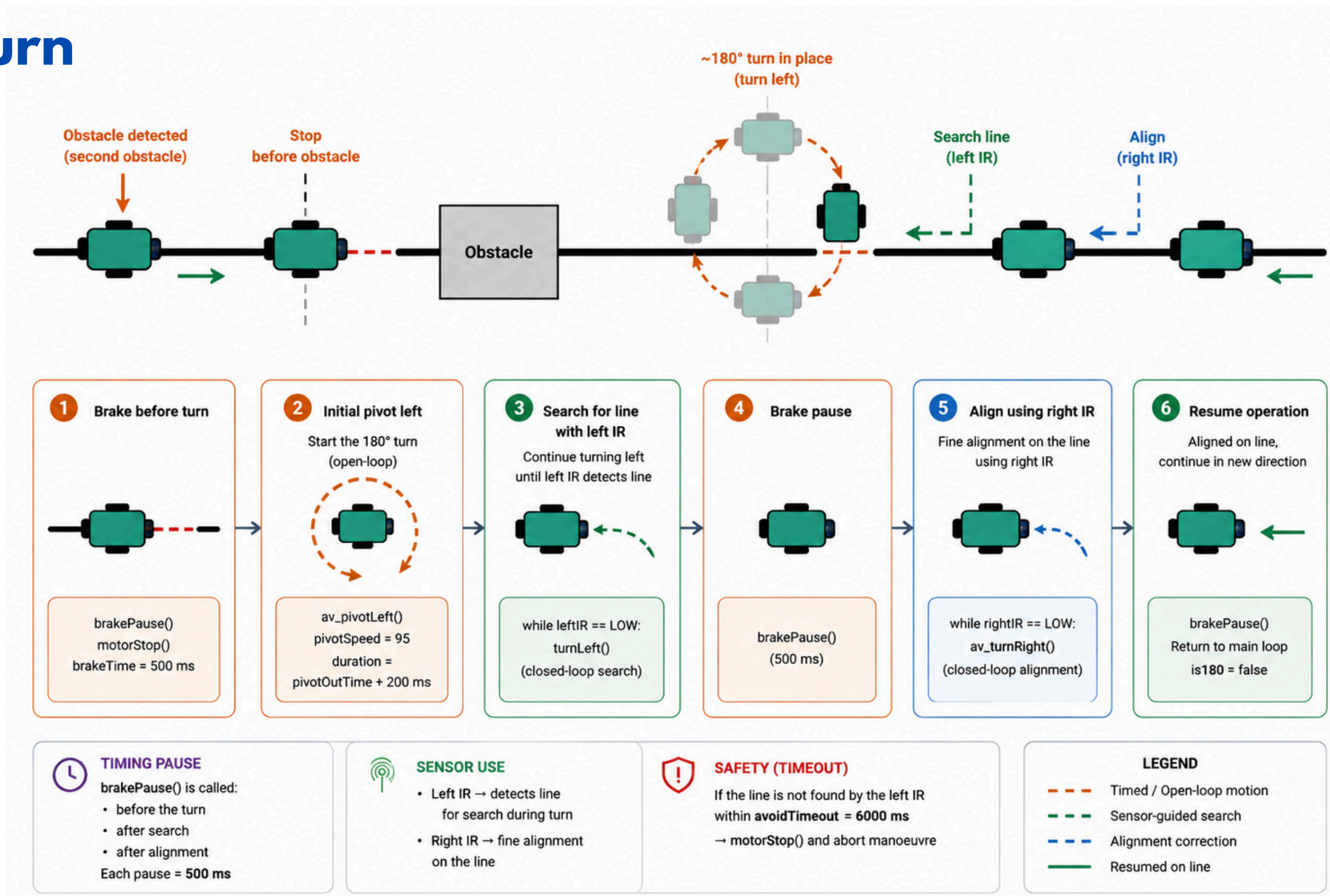


SOFTWARE & CONTROL LOGIC (3/3)

Slide 12



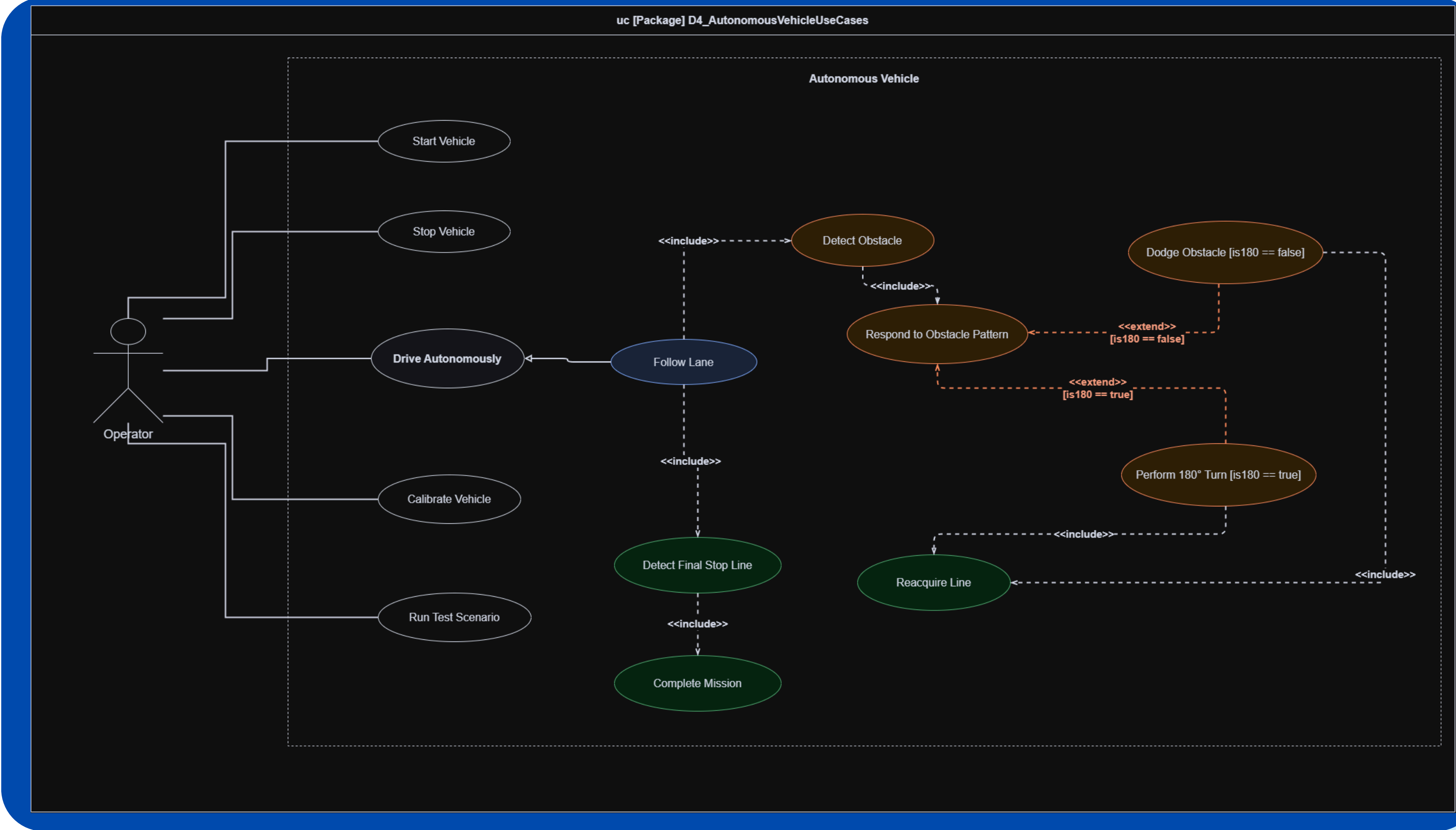
• 180° turn



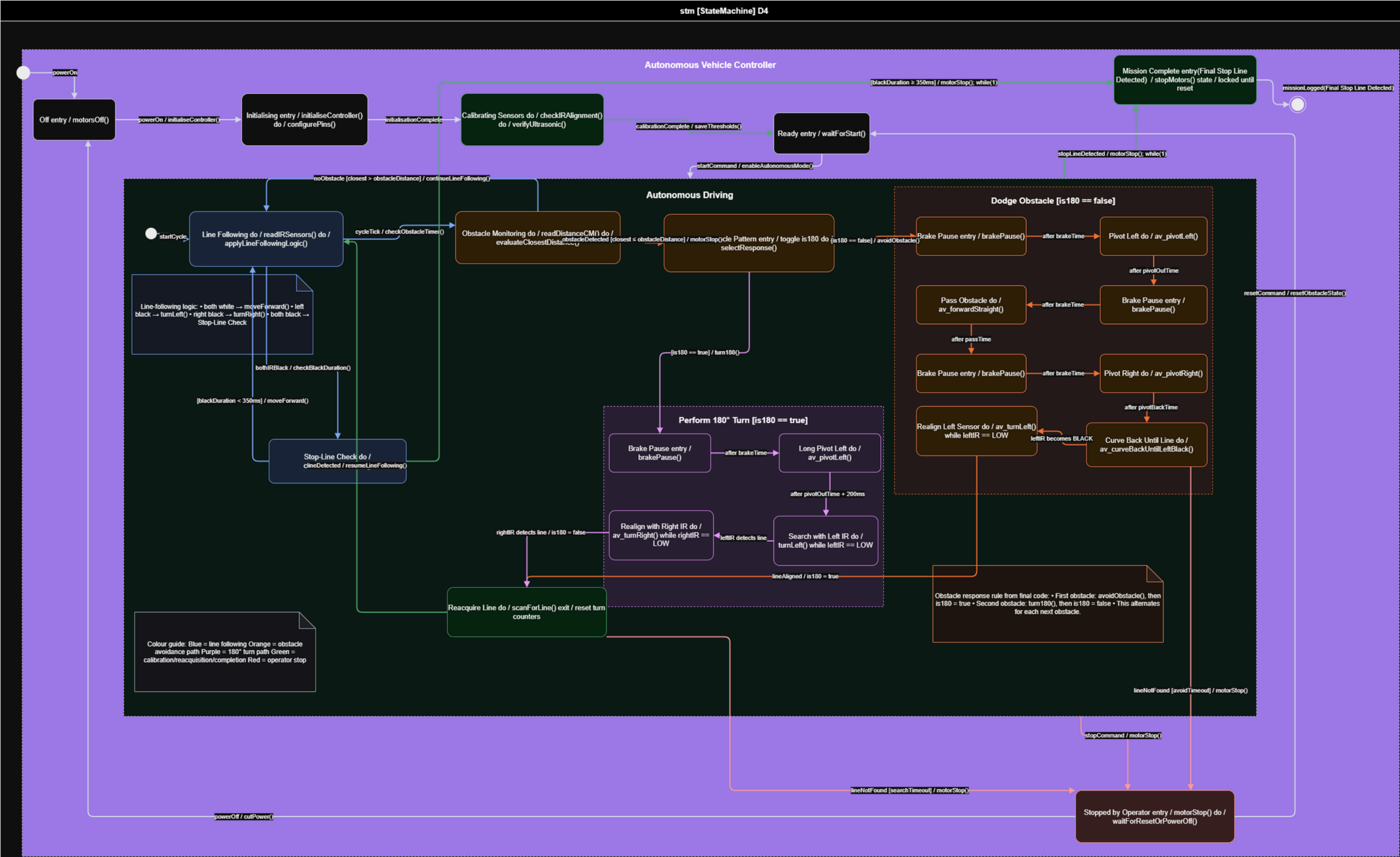
UML DIAGRAMS

Presented by ;

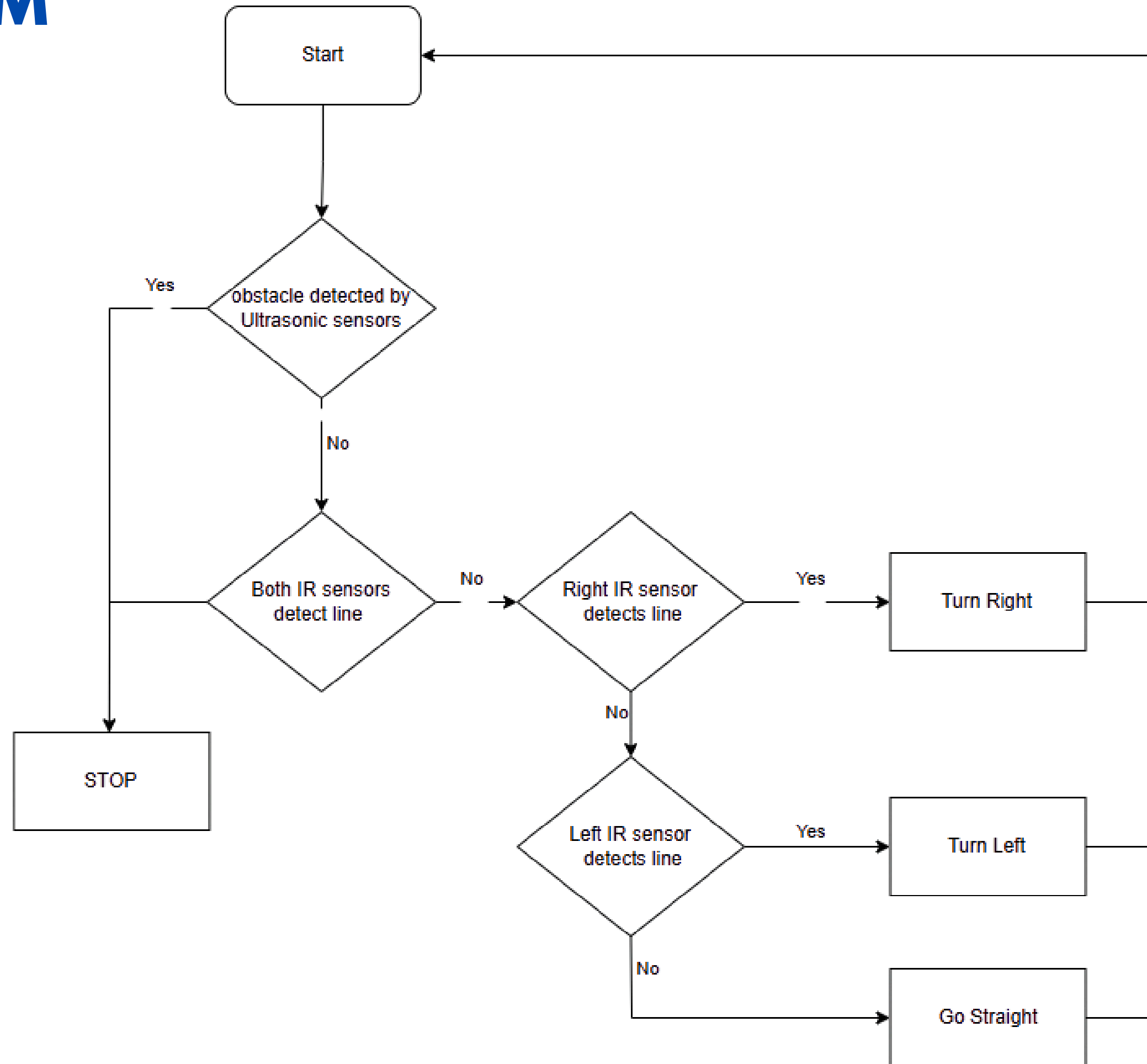
- Dodly Forbi
- Ye-Htet_Kyaw
- Gabriel Tchakunte
- Abdelmalek Abraham



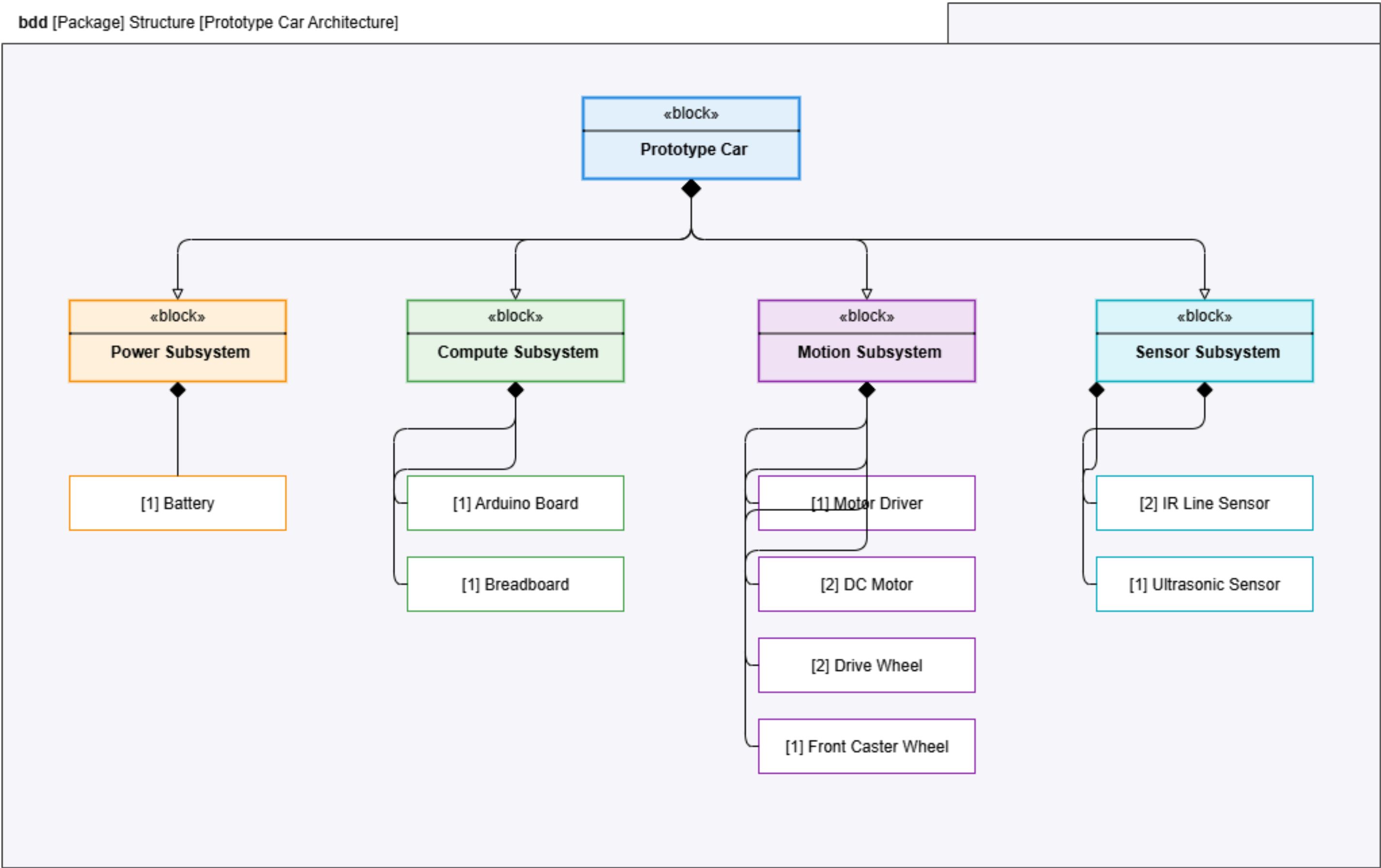
STATE MACHINE DIAGRAM



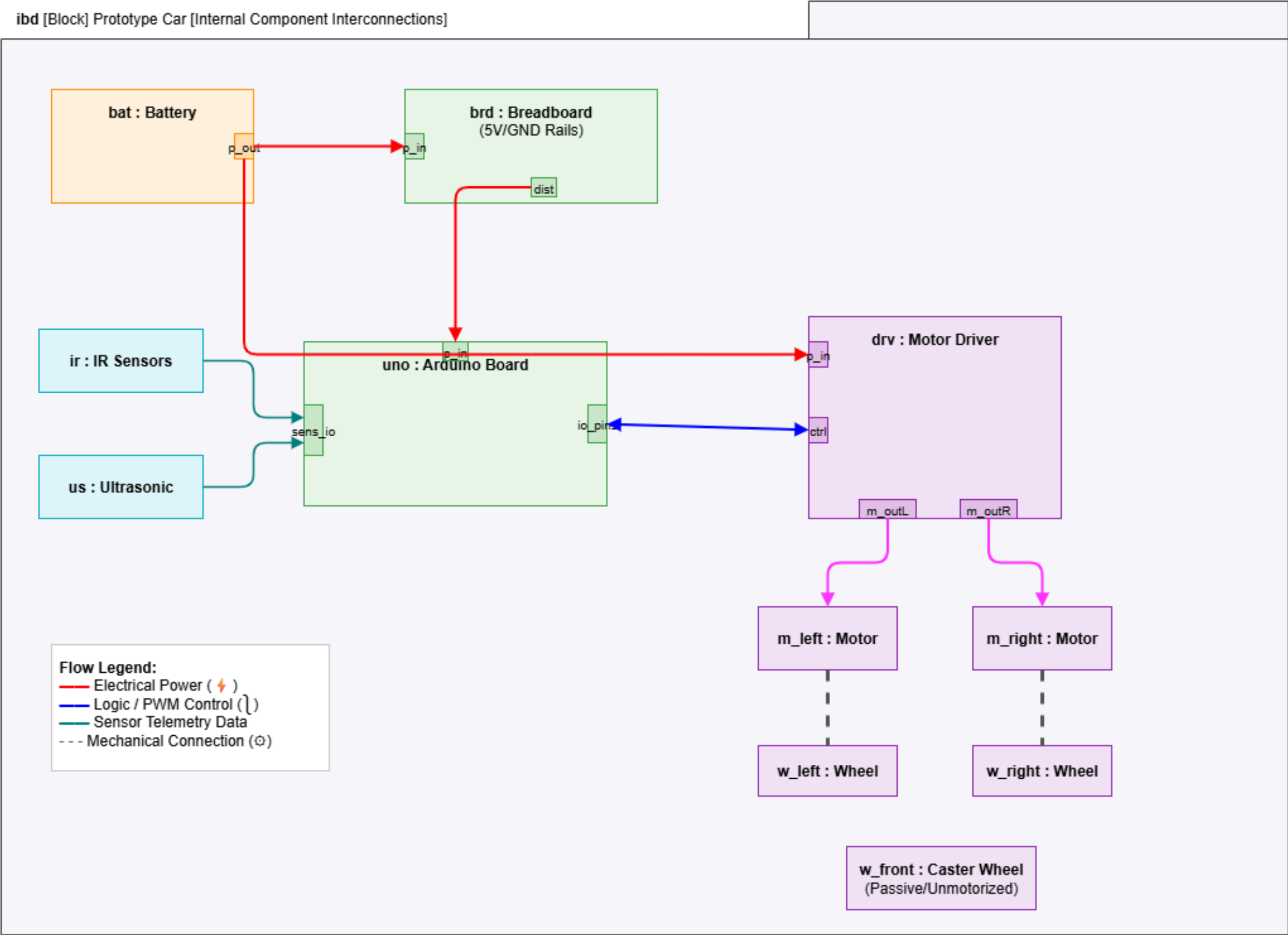
ACTIVITY DIAGRAM



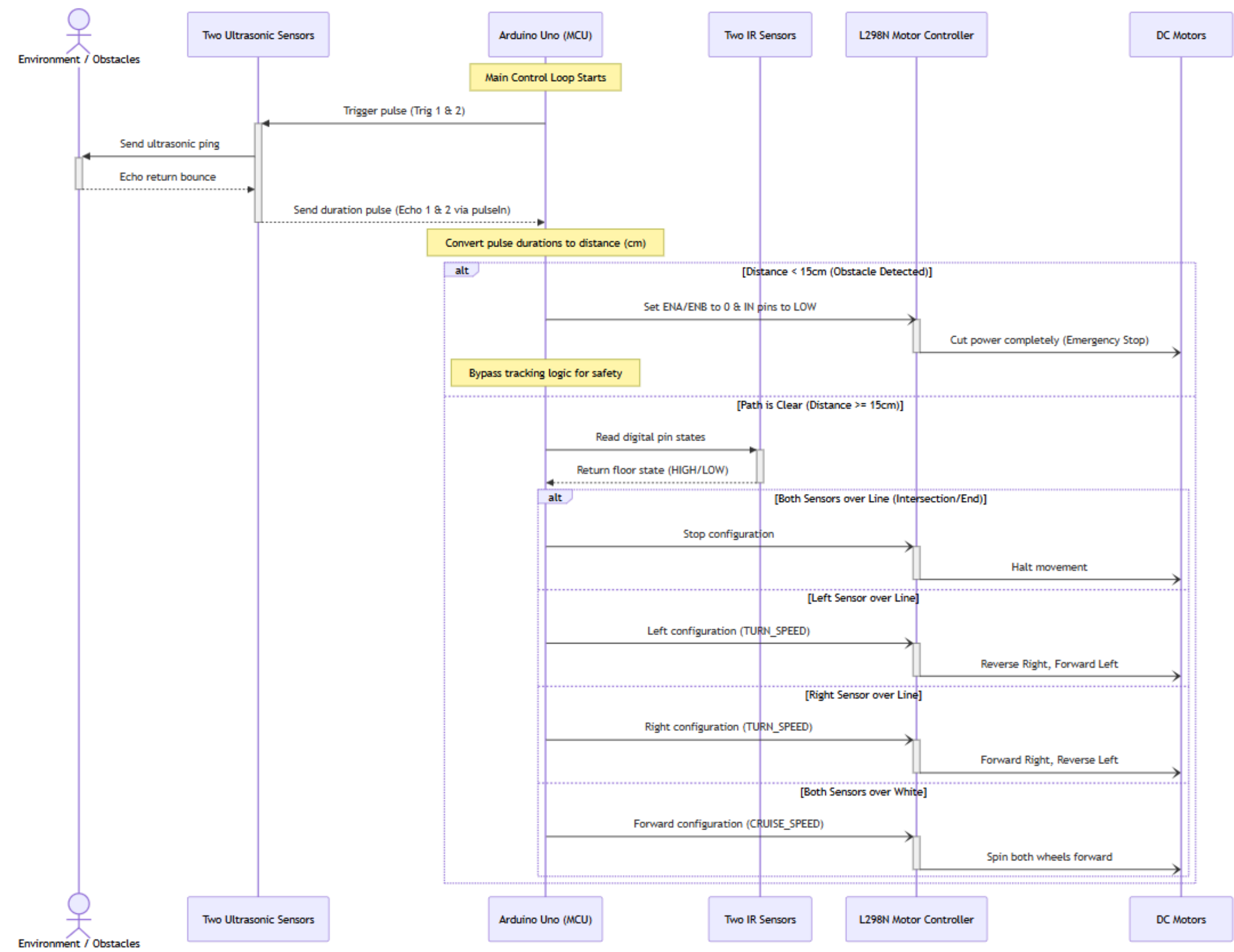
PACKAGE DIAGRAM



BLOCK COMPONENT DIAGRAM



SEQUENCE DIAGRAM



TESTING & RESULTS

Test Photo #1

Test Photo #2

Test Photo #3

TESTING & RESULTS (2/2)

RESULTS

- **Motor completed track in 48secs without obstacle manoeuvring**
- **Static friction required kick-start PWM.**
- **Physical testing outperformed simulation.**
- **Sensor placement mattered more than expected.**
- **Mechanical robustness required multiple redesigns.**

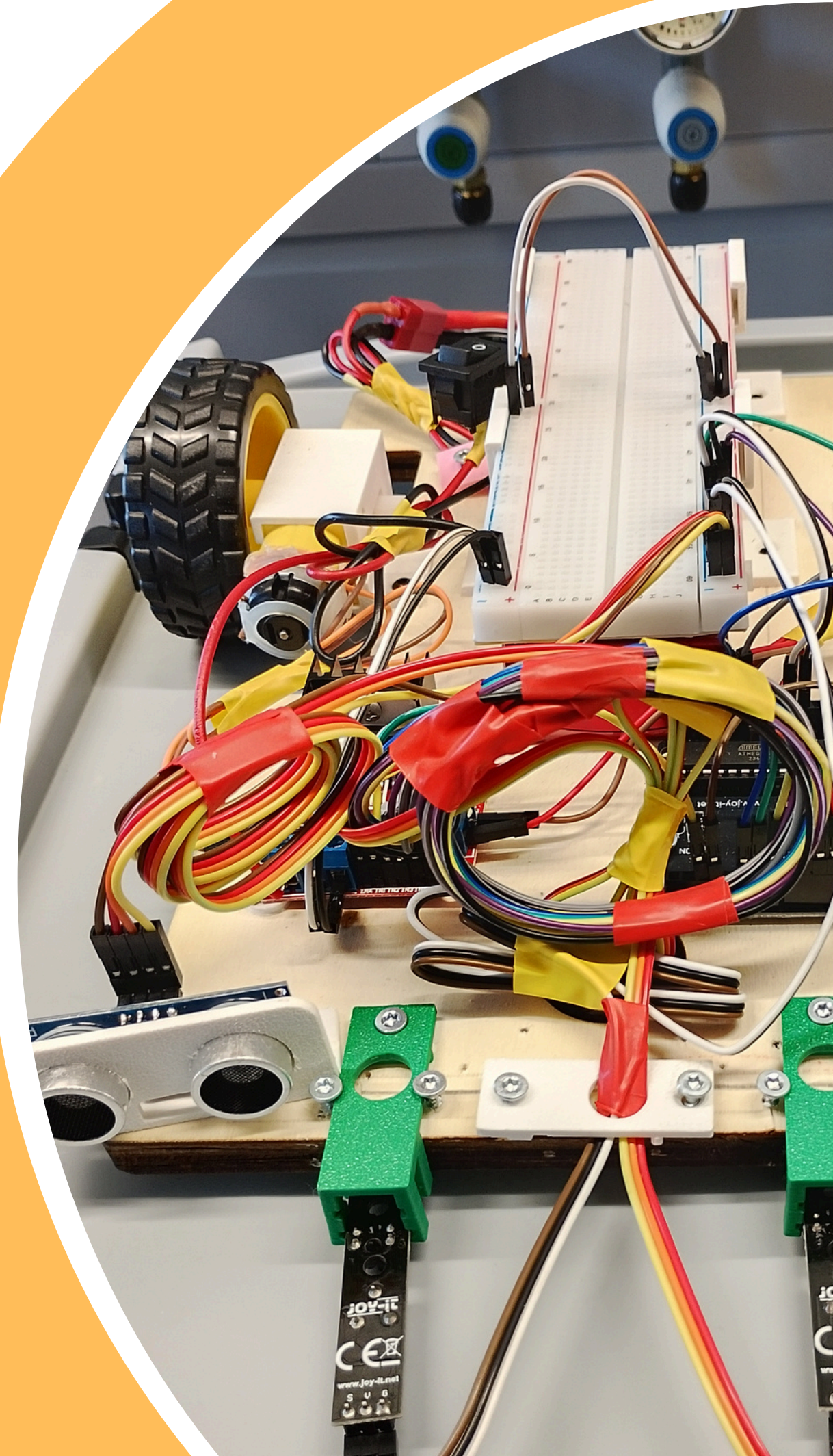
CONCLUSIONS

ACHIEVEMENTS

- ✓ **Met all objectives**
- ✓ **Used only project-kit hardware**
- ✓ **Reliable autonomous behaviour**
- ✓ **Complete documentation**
- ✓ **Team Collaboration on Github**

FUTURE WORK

- **PID line following**
- **Wheel encoders**
- **Adaptive IR calibration**
- **Vision-based navigation**



Thank You

for your keen attention.

Questions , comments ?

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